

8/ Laptops and Learning, again Refer to the data given in Exercise 16 (Section 12.3). The MINITAB printout is reproduced here.

Regression Analysis: y versus x

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	3254.03	3254.03	56.05	0.000
Error	18	1044.92	58.05		
Total	19	4298.95			

Model Summary

S	R-sq	R-sq(adj)
7.61912	75.69%	74.34%

Coefficients

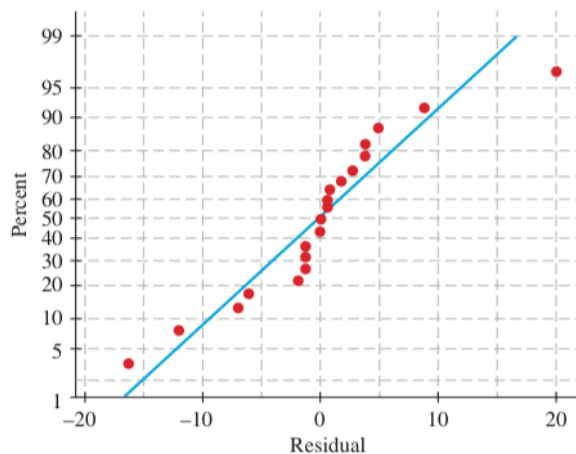
Term	Coef	SE Coef	T-Value	P-Value
Constant	-26.8	14.8	-1.82	0.086
x	1.262	0.169	7.49	0.000

Regression Equation

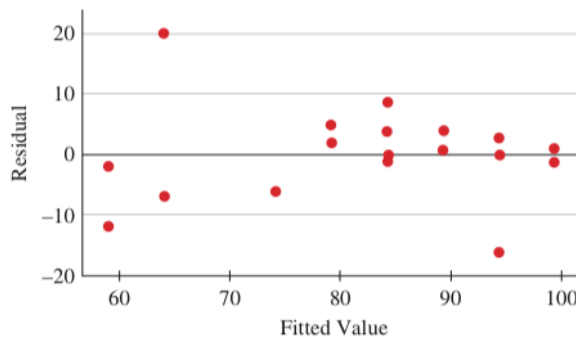
$$y = -26.8 + 1.262x$$

- What assumptions must be made about the distribution of the random error ε ?
- What is the best estimate of σ^2 , the variance of the random error ε ?
- Use the diagnostic plots for these data to comment on the validity of the regression assumptions.

Normal Probability Plot
(response is y)



Versus Fits
(response is y)



a. normal distribution:

mean = 0, common variance σ^2

independent x

b. $MSE = 58.05$

c. normality: $\checkmark$ 大部份分佈在直线附近

linear model: $\checkmark$

equal variance: $\checkmark$